CO4 – AT2 INDUSTRY PROBLEM SOLVING TASK

Answered Questions

The answers below follow the exact conditions, values, and tasks given in the uploaded Industry Problem Solving Task. No additional criteria have been introduced.

1. Autonomous Vehicle Vision System Selection

Question: An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions: If accuracy < 90% → system fails; if latency > 70 ms → unacceptable; the vehicle supports moderate computational resources. Methods available: Viola-Jones: 82% accuracy, 30 ms latency, low computation; YOLO: 93% accuracy, 60 ms latency, moderate computation; Deep Learning: 97% accuracy, 90 ms latency, high computation. Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

Method	Accuracy	Latency	Computation	Meets All Constraints?
Viola-Jones	82%	30 ms	Low	No – accuracy < 90%
YOLO	93%	60 ms	Moderate	Yes
Deep Learning	97%	90 ms	High	No – latency > 70 ms and computation exceeds moderate supp

Answer: Viola-Jones has acceptable latency at 30 ms and low computational requirements, but its accuracy is only 82%. Since the condition states that accuracy below 90% causes the system to fail, Viola-Jones is rejected.

YOLO provides 93% accuracy, which is above the required 90%. Its latency is 60 ms, which is below the maximum acceptable 70 ms, and its computation requirement is moderate, matching the vehicle's available computational resources. Therefore, YOLO satisfies all three constraints.

Deep Learning provides the highest accuracy at 97%, but its latency is 90 ms, exceeding the 70 ms limit. It also requires high computation while the vehicle supports only moderate computation. Therefore, Deep Learning is rejected.

Final decision: YOLO. YOLO is the only method that satisfies every stated constraint: accuracy ≥ 90%, latency ≤ 70 ms, and moderate computational requirement. It therefore provides the logically correct choice for the autonomous vehicle detection system under the specified conditions.

2. Smart Factory Product Inspection Selection

Question: A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions: If accuracy < 95% → defects may be missed; if processing time > 50 ms → production is affected; hardware supports moderate computation. Methods available: Traditional Image Analysis: 91% accuracy, 25 ms, low computation; YOLO: 96% accuracy, 45 ms, moderate computation; Deep Learning: 98% accuracy, 75 ms, high computation. Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.

Method	Accuracy	Processing Time	Computation	Meets All Constraints?
Traditional Image Analysis	91%	25 ms	Low	No – accuracy < 95%
YOLO	96%	45 ms	Moderate	Yes
Deep Learning	98%	75 ms	High	No – processing time > 50 ms and computation exceed

Answer: Traditional Image Analysis has a fast processing time of 25 ms and low computational requirements, but its accuracy is only 91%. Because the condition requires at least 95% accuracy, it is rejected even though its speed and computation are suitable.

YOLO provides 96% accuracy, which satisfies the minimum 95% requirement. Its processing time is 45 ms, which is below the 50 ms limit, and it requires moderate computation, matching the available hardware. Therefore, YOLO satisfies all three constraints.

Deep Learning gives the highest accuracy at 98%, but its processing time is 75 ms, exceeding the 50 ms maximum. It also requires high computation while the hardware supports only moderate computation. Therefore, it is rejected.

Final decision: YOLO. YOLO is the only method satisfying accuracy ≥ 95%, processing time ≤ 50 ms, and moderate computational requirements. It is therefore the most suitable method for the real-time production-line inspection system under the stated constraints.

Overall Conclusion

For both industry problems, YOLO is the only option that satisfies every stated constraint. In the autonomous vehicle case, it achieves 93% accuracy, 60 ms latency, and moderate computation. In the smart factory case, it achieves 96% accuracy, 45 ms processing time, and moderate computation. The other methods each violate at least one mandatory condition.